

RAJ BHARATI

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Portfolio: rajbharati2705.github.io/portfolio | LinkedIn: linkedin.com/in/raj-bharati | GitHub: github.com/rajbharati2705

PROFESSIONAL SUMMARY

Software Engineer with 2+ years of experience building scalable backend and full-stack applications using Node.js, TypeScript, AWS, and SQL databases. Experienced in designing distributed systems, event-driven architectures, real-time communication, and cloud-native applications. Proven track record of improving system performance by 45%, reducing operational costs by 70%, and building highly available APIs supporting 1000+ concurrent connections.

CORE COMPETENCIES

Backend Development • Distributed Systems • Event-Driven Architecture • Real-Time Systems • Microservices Architecture • System Design • API Development & Optimization • Database Performance Tuning • Cloud Infrastructure (AWS) • CI/CD & DevOps • Performance Engineering • Scalability & High Availability • Fault Tolerance • Asynchronous Processing • Horizontal Scaling

EXPERIENCE

Software Engineer | Backend & Full Stack Developer

DSmart Systems Private Limited | Oncare360 Healthcare Platform

Jan 2024 – Present

Healthcare SaaS platform serving medical professionals and patients with real-time communication features

- Designed and implemented Redis Pub/Sub-based distributed event propagation system to synchronize WebSocket events across AWS ECS containers, enabling horizontally scalable real-time systems supporting 1000+ concurrent connections.
- Designed and built CI/CD pipelines using GitHub Actions for automated testing and deployment to AWS EC2/ECS, reducing deployment time by 40% and improving release reliability.
- Automated database migrations using Liquibase integrated with pull request workflows, ensuring zero-downtime deployments and database schema consistency across environments.
- Led migration of application monitoring from Datadog to Grafana and Prometheus, reducing operational costs by 70% while improving system observability, alerting capabilities, and monitoring reliability.
- Implemented Redis caching strategies and API optimization techniques, reducing API latency by 45% and database load by 60% for high-frequency requests, improving overall system throughput.
- Automated event-driven billing workflows using AWS SQS and Lambda for asynchronous processing, improving billing efficiency by 60% through queue-based task execution and fault-tolerant message handling.
- Optimized MySQL performance through stored procedures, indexing strategies, query tuning, and connection pooling, reducing query response times by 35%.
- Designed and implemented RESTful APIs, webhooks, and scheduled cron jobs for scalable backend workflows supporting healthcare operations.
- Implemented Docker containerization with multi-stage builds and created comprehensive Swagger/OpenAPI documentation for API endpoints.
- Established Jest unit testing framework achieving 80% code coverage, improving code maintainability and reducing production bugs by 50%.

PROJECTS

API Monitoring & Observability Platform

GitHub: github.com/rajbharati2705/API-Monitoring-Service

- Designed and built a production-grade API monitoring platform capable of tracking uptime, latency, and system health in real time using event-driven architecture with RabbitMQ and WebSockets, achieving 99.9% uptime.

- Implemented fault-tolerant asynchronous processing using RabbitMQ message queues with retry mechanisms, dead-letter queues (DLQ), and circuit breaker patterns for reliable error handling and system resilience.
- Developed high-throughput ingestion pipeline with Express.js background workers processing 10,000+ API metrics per minute with sub-100ms latency.
- Built a scalable data layer using MongoDB for raw event storage (time-series data) and PostgreSQL for aggregated analytics, enabling real-time dashboards and historical trend analysis with 50% faster query performance.

Tech Stack: Node.js, Express.js, RabbitMQ, MongoDB, PostgreSQL, WebSockets, Docker, CI/CD, REST APIs, System Design

SkyScanner - Flight Search and Booking Backend

GitHub: github.com/rajbharati2705/Flight-booking-service

- Developed a highly scalable backend for flight search and booking using microservices architecture with service-oriented design patterns, supporting 500+ concurrent search requests.
- Implemented robust error handling, database indexing optimization, and deadlock resolution mechanisms, reducing query response time by 50% and ensuring 99.5% system reliability.
- Utilized Sequelize ORM with MySQL to ensure ACID compliance, data integrity, and optimal database performance through efficient transaction management and query optimization.

Tech Stack: Node.js, Express.js, MySQL, Sequelize ORM, Microservices, REST APIs

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Java (Spring Boot - Intermediate), SQL, HTML, CSS

Backend & Architecture: Node.js, Express.js, REST APIs, GraphQL, WebSockets, Microservices Architecture, Event-Driven Architecture, Message Queues (RabbitMQ, AWS SQS), Distributed Systems, Server-Side Development, System Design, Asynchronous Processing

Frontend: React, Angular

Databases & Caching: MySQL, PostgreSQL, MongoDB (NoSQL), Redis (Caching & Pub/Sub), Database Optimization, Indexing, Query Tuning

ORMs & ODMs: Prisma, Sequelize, Mongoose

Cloud & DevOps: AWS (EC2, ECS, Lambda, SQS, S3, ECR, Auto Scaling, CloudWatch), Docker, Kubernetes (Basic), GitHub Actions, CI/CD Pipelines, Continuous Integration, Continuous Deployment, Infrastructure as Code, Grafana, Prometheus, Liquibase

Testing & Tools: Jest, Mocha, Postman, Swagger/OpenAPI, Git, GitHub, VS Code, Turborepo, npm, yarn

Methodologies: Agile/Scrum, Test-Driven Development (TDD), Code Review, API-First Design, Scalability & High Availability, Fault Tolerance, Performance Optimization

EDUCATION

Nalla Malla Reddy Engineering College

Hyderabad, India

Bachelor of Technology (BTech) in Electronics and Communication Engineering (ECE) | CGPA: 7.34/10.0 2019 - 2023